Time Series Homework 5

Matvei Fedortsov

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1 Task 1

1.1 Introduction

Volatility models such as ARCH (Autoregressive Conditional Heteroskedasticity) and GARCH (Generalized ARCH) are widely used in financial econometrics to model time-varying variance. In this exercise, we generate data from known ARCH(4) and GARCH(1,1) processes, then estimate the models using the same specification. The goal is to assess the finite-sample performance of the maximum likelihood estimator and to observe the typical behaviour of the estimates as the sample size changes.

1.2 Data Generation

All series are generated with i.i.d. standard normal innovations $z_t \sim \mathcal{N}(0,1)$. A burn-in period of 1,000 observations is used to eliminate start-up effects.

1.3 ARCH(4) Model

The true ARCH(4) model is given by

$$\begin{aligned}\varepsilon_t &= \sigma_t z_t,\\ \sigma_t^2 &= 1 + 0.3\varepsilon_{t-1}^2 + 0.25\varepsilon_{t-2}^2 + 0.15\varepsilon_{t-3}^2 + 0.1\varepsilon_{t-4}^2.\end{aligned}$$

The unconditional variance is $E[\varepsilon_t^2] = 1/(1 - 0.3 - 0.25 - 0.15 - 0.1) = 5$.

1.4 GARCH(1,1) Model

The true GARCH(1,1) model is

$$\begin{aligned}\varepsilon_t &= \sigma_t z_t,\\ \sigma_t^2 &= 0.1 + 0.2\varepsilon_{t-1}^2 + 0.7\sigma_{t-1}^2.\end{aligned}$$

Persistence is $\alpha + \beta = 0.9$, and unconditional variance is $0.1/(1 - 0.9) = 1$.

1.4.1 ARCH(4) Estimation Results

The ARCH(4) model was estimated using zero mean specification. Table 1 summarises the true parameters and the estimates (with standard errors in parentheses) for the three sample sizes.

For the largest sample ($N = 10000$) the estimated ARCH(4) equation is

$$\hat{\sigma}_t^2 = 1.032 + 0.298\,\varepsilon_{t-1}^2 + 0.239\,\varepsilon_{t-2}^2 + 0.130\,\varepsilon_{t-3}^2 + 0.112\,\varepsilon_{t-4}^2. \tag{1}$$

All coefficients are statistically significant (p-values < 0.001) and close to the true values. The sum of the ARCH coefficients is 0.779, slightly below the true 0.8, yielding an estimated unconditional variance of 4.675 compared to the true 5.000.

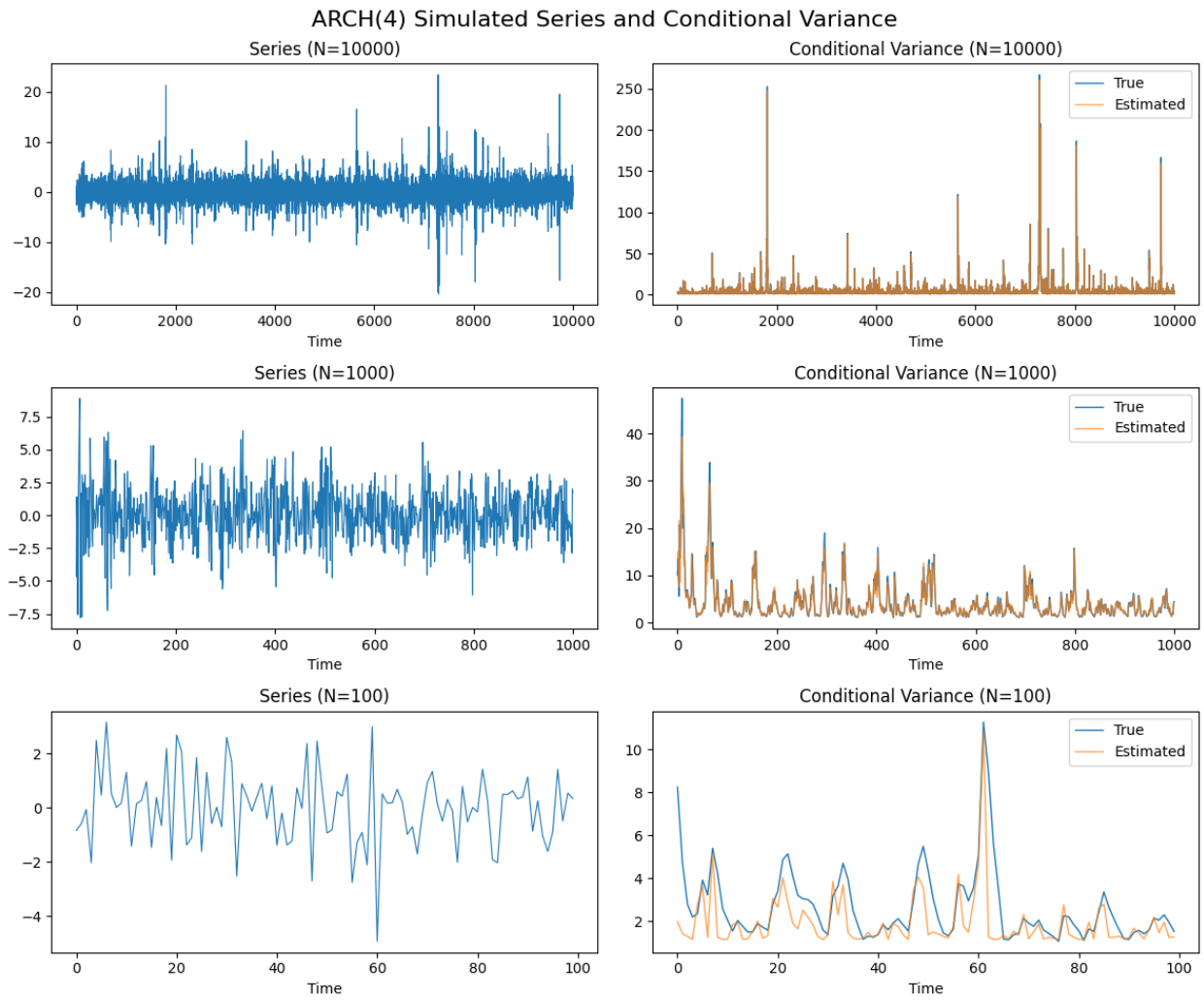

Figure 1: ARCH plots

Table 1: ARCH(4) estimation results. Standard errors are shown in parentheses.

Parameter	True	$N = 10000$	$N = 1000$	$N = 100$
ω	1.000	1.032 (0.041)	1.136 (0.132)	1.157 (0.394)
α_1	0.300	0.298 (0.017)	0.308 (0.057)	0.398 (0.159)
α_2	0.250	0.239 (0.015)	0.117 (0.035)	0.000 (0.103)
α_3	0.150	0.130 (0.014)	0.184 (0.041)	0.000 (0.085)
α_4	0.100	0.112 (0.014)	0.115 (0.039)	0.000 (0.217)
$\sum \alpha$	0.800	0.779	0.723	0.398
Uncond. var.	5.000	4.675	4.107	1.922
Sample mean	0.000	0.031	0.108	-0.015
Sample var.	5.000	4.919	3.946	1.905
Sample skew	0.000	-0.013	-0.054	-0.245

As the sample size decreases, the estimates become less precise. For $N = 1000$, the second lag is underestimated (0.117 vs. 0.25) while the third lag is overestimated (0.184 vs. 0.15). For $N = 100$, only the first lag remains significant; the others are estimated near zero with large standard errors. The unconditional variance estimate drops to 1.922, far from the true 5, and the sample skewness deviates noticeably from zero.

1.5 GARCH(1,1) Estimation Results

The GARCH(1,1) model was estimated with a zero mean. Table 6 presents the results.

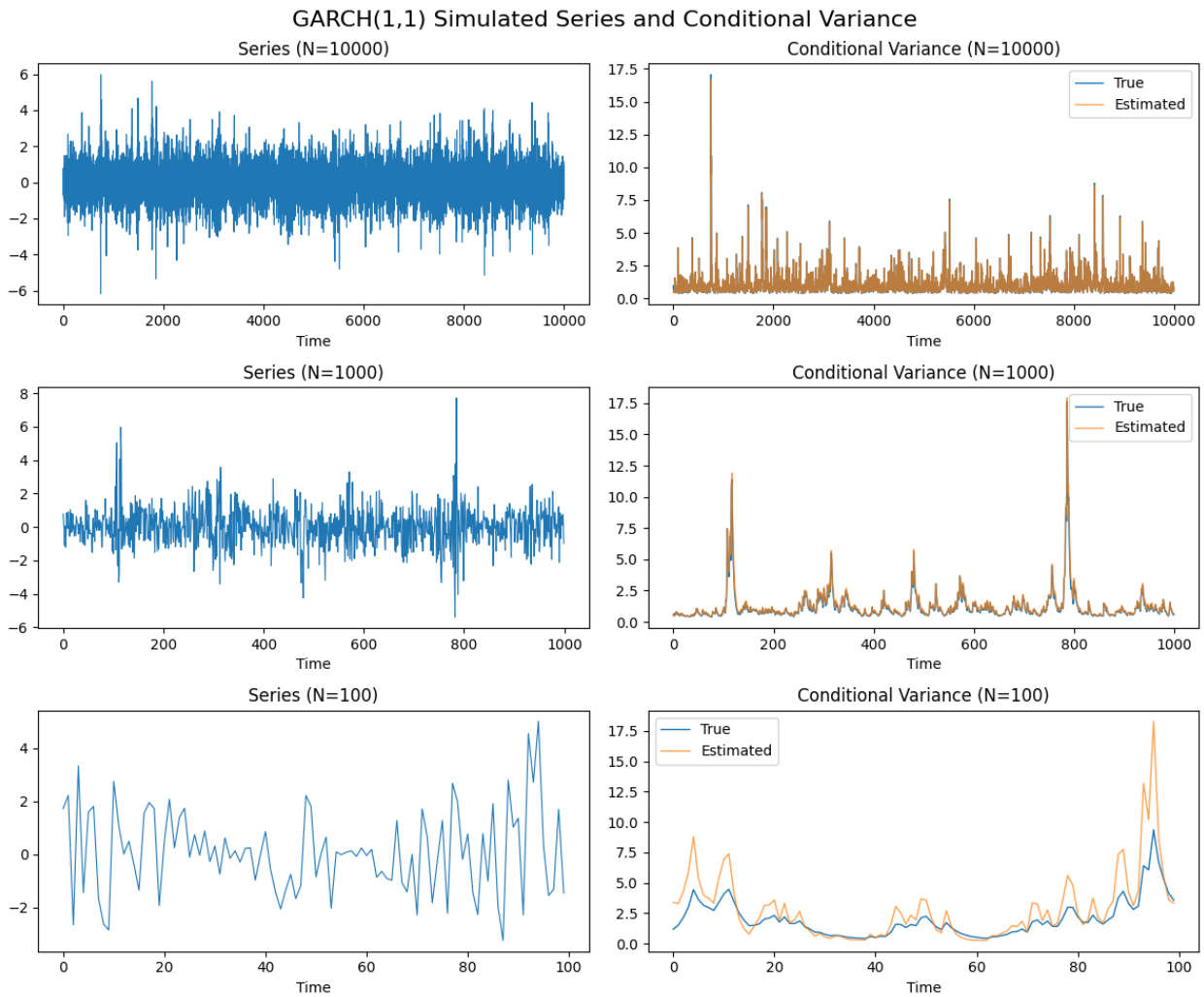

Figure 2: GARCH plots

Table 2: GARCH(1,1) estimation results. Standard errors are shown in parentheses.

Parameter	True	$N = 10000$	$N = 1000$	$N = 100$
ω	0.100	0.114 (0.010)	0.092 (0.025)	0.121 (0.078)
α	0.200	0.197 (0.011)	0.194 (0.030)	0.534 (0.149)
β	0.700	0.694 (0.017)	0.735 (0.035)	0.466 (0.120)
$\alpha + \beta$	0.900	0.891	0.929	1.000
Uncond. var.	1.000	1.045	1.296	– (not defined)
Sample mean	0.000	-0.007	0.020	0.099
Sample var.	1.000	1.042	1.328	2.581
Sample skew	0.000	0.020	0.383	0.378

For $N = 10000$, the estimated GARCH(1,1) equation is

$$\hat{\sigma}_t^2 = 0.114 + 0.197\,\varepsilon_{t-1}^2 + 0.694\,\sigma_{t-1}^2. \tag{2}$$

All coefficients are highly significant and close to the true values. The persistence is estimated at 0.891, slightly below the true 0.9, giving an unconditional variance of 1.045 (true 1).

With $N = 1000$, the estimates remain reasonable, though the persistence is a bit higher (0.929). The unconditional variance is overestimated at 1.296. For $N = 100$, the estimate of α becomes 0.534 (much too high) and β becomes 0.466 (too low). The persistence reaches exactly 1.000, implying an integrated GARCH process and an undefined unconditional variance. The sample variance is also inflated (2.581 instead of 1). Skewness is noticeably positive for the two smaller samples.

2 Task 7

2.1 Introduction

The objective of this study is to model the volatility of the MOEX stock index using ARMA-GARCH type models. Daily closing prices are taken from the `Russtockdaily.csv` file, covering observations from index 3516 to 4441 (926 data points). Log returns are computed to achieve stationarity. After a preliminary analysis, we estimate a range of volatility models (GARCH, EGARCH, GJR-GARCH, APARCH, FIGARCH, PowerGARCH) combined with low-order AR mean specifications. The models are compared on the basis of the Akaike information criterion (AIC), the Ljung–Box test for autocorrelation in the residuals and squared residuals, and the number of parameters (restricted to at most six). The two best performing specifications are then used to generate forecasts of both the conditional mean and the conditional volatility.

2.2 Data Description

The raw price series exhibits a clear upward trend, confirming non-stationarity. Figure 3 shows the evolution of the MOEX price. To obtain a stationary series we compute log returns:

$$r_t = \ln(P_t) - \ln(P_{t-1}),$$

where P_t is the closing price on day t. The returns series, plotted in Figure 4, appears stable around zero with volatility clustering.

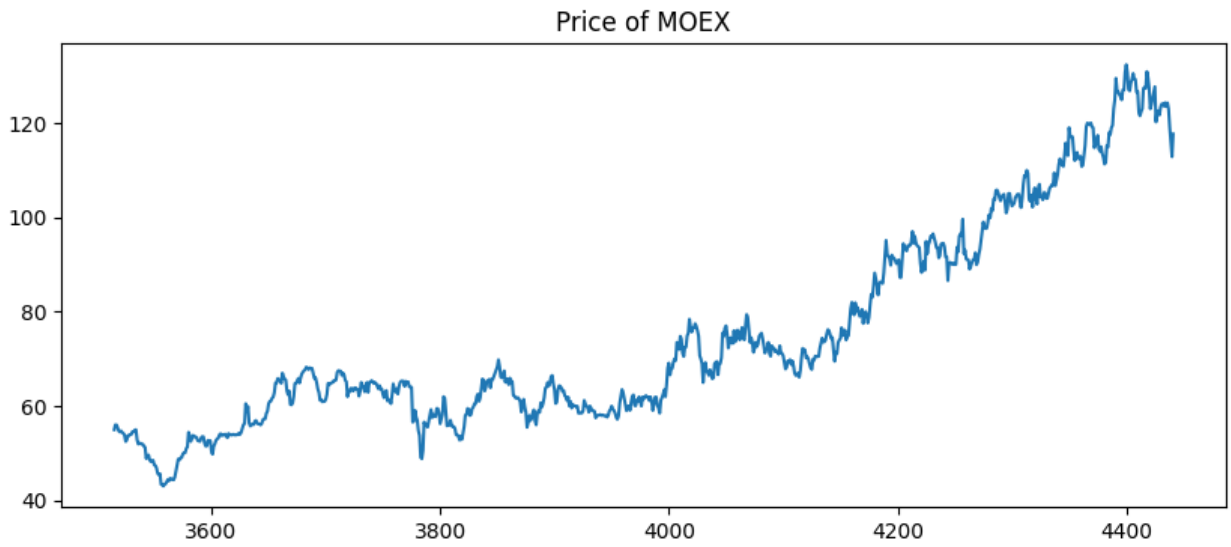

Figure 3: MOEX daily closing prices.

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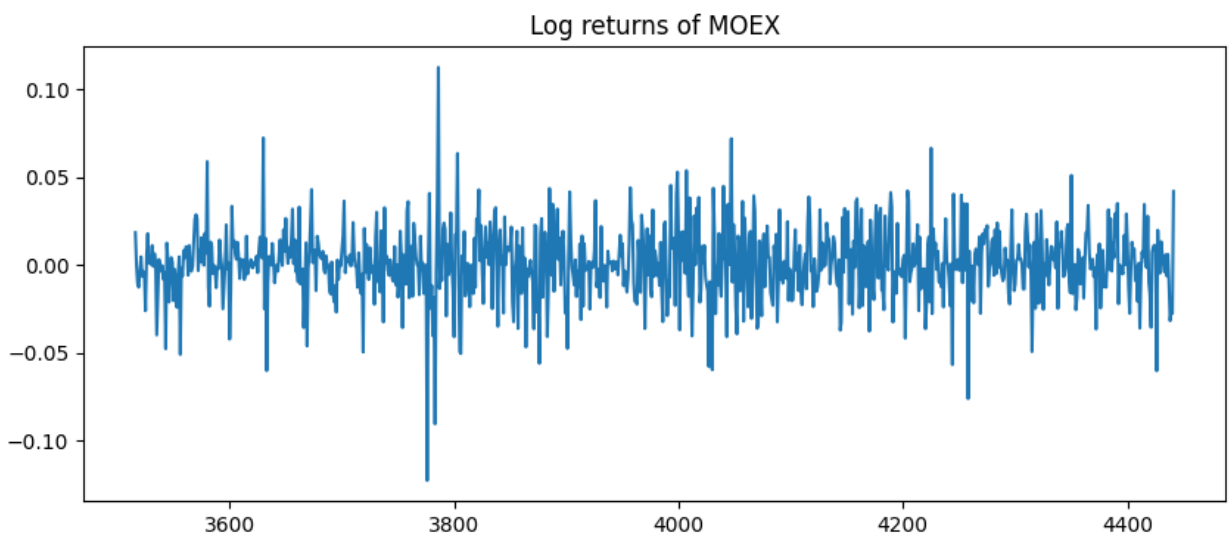

Figure 4: MOEX daily log returns.

Figure 5: Time series plots.

2.3 Stationarity Tests

Three unit-root tests are applied to both the price and the returns: the Augmented Dickey–Fuller (ADF) test, the Phillips–Perron (PP) test, and the KPSS test. The results are summarised in Table 3.

Table 3: Stationarity test results.

Series	Test	Statistic	p-value	Conclusion
Price	ADF	-0.3612	0.9164	Non-stationary
	PP	-0.3612	0.9164	Non-stationary
	KPSS	4.0648	0.0100	Non-stationary
Returns	ADF	-19.2034	0.0000	Stationary
	PP	-29.9488	0.0000	Stationary
	KPSS	0.0767	0.1000	Stationary

All tests unanimously indicate that the price series is non-stationary, whereas the log returns are stationary. Hence we proceed with the returns for further modelling.

2.4 ARMA Order Selection

An automatic ARIMA selection (using the `auto_arima` function) is performed on the returns. The chosen model is an ARIMA(0,0,0) with a constant, i.e. a simple white noise process. This suggests that the conditional mean of the returns contains no significant autoregressive or moving average components. Consequently, we will only consider AR(0) (constant mean) specifications for the mean equation in the GARCH family models.

2.5 ARMA-GARCH Framework

For a time series of returns r_t, the general ARMA(m, n)-GARCH(p, q) model is written as

$$r_t = \mu + \sum_{i=1}^{m} \phi_i r_{t-i} + \sum_{j=1}^{n} \theta_j \varepsilon_{t-j} + \varepsilon_t,$$

with $\varepsilon_t = \sigma_t z_t$, $z_t \sim$ i.i.d.$(0, 1)$, and σ_t^2 follows a GARCH-type process. In this study we restrict the mean equation to a constant (μ) because the preliminary analysis indicates no ARMA structure. The distribution of z_t is taken as the standardised Student's t to capture heavy tails.

2.6 Volatility Specifications

We consider several volatility models, each with a maximum of six parameters (including the distributional parameter). The main specifications are:

- **GARCH(p, q)** (Bollerslev, 1986):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2,$$

  with $\omega > 0,\ \alpha_i, \beta_j \geq 0$.

- **EGARCH(p, q)** (Nelson, 1991):

$$\ln(\sigma_t^2) = \omega + \sum_{i=1}^{p} (\alpha_i |z_{t-i}| + \gamma_i z_{t-i}) + \sum_{j=1}^{q} \beta_j \ln(\sigma_{t-j}^2),$$

  where $z_t = \varepsilon_t / \sigma_t$. This model allows for asymmetric effects of shocks.

- **GJR-GARCH(p, q)** (Glosten, Jagannathan & Runkle, 1993), also known as TGARCH:

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \left(\alpha_i \varepsilon_{t-i}^2 + \gamma_i I_{t-i} \varepsilon_{t-i}^2\right) + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2,$$

  with $I_{t-i} = 1$ if $\varepsilon_{t-i} < 0$ and 0 otherwise. The leverage effect is captured by $\gamma_i > 0$.

- **APARCH(p, q)** (Ding, Granger & Engle, 1993):

$$\sigma_t^\delta = \omega + \sum_{i=1}^{p} \alpha_i (|\varepsilon_{t-i}| - \gamma_i \varepsilon_{t-i})^\delta + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^\delta,$$

  where $\delta > 0$ and $|\gamma_i| \leq 1$. The symmetric PowerGARCH is a special case with $\gamma_i = 0$.

- **FIGARCH(p, d, q)** (Baillie, Bollerslev & Mikkelsen, 1996):

$$\phi(L)(1-L)^d \varepsilon_t^2 = \omega + [1 - \beta(L)]\nu_t,$$

  where $\nu_t = \varepsilon_t^2 - \sigma_t^2$ and d is the fractional differencing parameter. This model captures long memory in volatility.

In addition, we considered IGARCH and GARCH-in-mean (GARCH-M) specifications, but they either exceeded the parameter limit or failed to converge for our data.

2.7 Model Selection Criteria

Models are evaluated based on:

- **Information criteria**: AIC and BIC (lower is better).
- **Ljung–Box test** on the standardised residuals and squared standardised residuals at lag 10. A p-value above 0.05 indicates no significant autocorrelation.
- **Number of parameters**: restricted to at most six to maintain parsimony.
- Convergence of the maximum likelihood estimation.

2.8 GARCH Model Comparison

We estimated all combinations of the volatility models with AR(0), AR(1) and AR(2) mean equations. Only models that converged and had at most six parameters are retained for the final comparison. Table 4 presents these models sorted by AIC. For models with AR(1) or AR(2) the Ljung–Box statistics could not be computed (possibly due to numerical issues), so we focus on the AR(0) specifications, which all pass the residual autocorrelation test ($p > 0.05$) but fail the squared residual test ($p \approx 0$), indicating that some volatility dynamics remain unexplained.

Table 4: Comparison of AR(0)-GARCH models with ≤ 6 parameters.

Model	Params	AIC	BIC	LogLik	LB(10) resid p	LB(10) sq. p
AR(0)-EGARCH(1,1)	6	-4731.5	-4702.5	2371.7	0.4989	0.0000
AR(0)-FIGARCH(1,1)	6	-4729.8	-4700.8	2370.9	0.4989	0.0000
AR(0)-PowerGARCH(1,1)	6	-4702.9	-4673.9	2357.4	0.4989	0.0000
AR(0)-GARCH(1,1)	5	-4701.1	-4676.9	2355.5	0.4989	0.0000
AR(0)-GARCH(1,2)	6	-4700.7	-4671.7	2356.4	0.4989	0.0000
AR(0)-GARCH(2,1)	6	-4699.9	-4670.9	2356.0	0.4989	0.0000
AR(0)-GJR-GARCH(1,1)	6	-4698.7	-4669.7	2355.3	0.4989	0.0000

2.9 Selection of the Two Best Models

Based on the lowest AIC and the need for a parsimonious representation, we select:

1. **AR(0)-EGARCH(1,1)** – it has the smallest AIC and captures asymmetric volatility responses.

2. **AR(0)-GARCH(1,1)** – the classic benchmark model, which also achieves a low AIC and is widely used in practice.

Although the FIGARCH model has a slightly better AIC, it is more complex and may be prone to overfitting; therefore we retain the more standard EGARCH and GARCH specifications. Both models are re-estimated with a Student's t distribution; the coefficient estimates are reported in Tables 5 and 6.

Table 5: AR(0)-EGARCH(1,1) with Student's t distribution.

Parameter	Coef.	Std. Err.	t-ratio	p-value
μ (mean)	0.000902	0.000516	1.748	0.080
ω	-0.0905	0.141	-0.642	0.521
α_1	0.0678	0.0640	1.059	0.289
γ_1	-0.0151	0.0219	-0.692	0.489
β_1	0.9875	0.0187	52.94	0.000
ν (dof)	4.2113	0.732	5.754	0.000

Table 6: AR(0)-GARCH(1,1) with Student's t distribution.

Parameter	Coef.	Std. Err.	t-ratio	p-value
μ (mean)	0.000935	0.000556	1.682	0.093
ω	4.00e-05	4.09e-06	9.784	0.000
α_1	0.0499	0.0217	2.297	0.022
β_1	0.8480	0.0233	36.43	0.000
ν (dof)	5.7176	2.883	1.983	0.047

In both models the degrees-of-freedom parameter ν is significantly greater than 2, confirming the need for a fat-tailed distribution. The EGARCH model shows an insignificant leverage term (γ_1), suggesting that asymmetry may not be strong in this series. The GARCH model satisfies the usual stationarity condition $\alpha_1 + \beta_1 < 1$.

3 Forecasting

We generate five-step-ahead forecasts of the conditional mean (returns) and conditional volatility (standard deviation) from each selected model. The forecasts are produced using a simulation method to account for the non-linear nature of EGARCH and the GARCH recursion. Table 7 reports the numerical values; Figure **??** displays the historical returns and volatility together with the forecasts.

Table 7: Five-step-ahead forecasts.

Horizon	EGARCH		GARCH	
	Return	Volatility	Return	Volatility
1	0.000902	0.02175	0.000935	0.02171
2	0.000902	0.02179	0.000935	0.02152
3	0.000902	0.02178	0.000935	0.02132
4	0.000902	0.02178	0.000935	0.02120
5	0.000902	0.02179	0.000935	0.02107

The point forecasts of the returns are constant because both models have a constant mean. The volatility forecasts from the EGARCH model remain almost flat, reflecting the high persistence estimated ($\beta_1 \approx 0.99$).

In contrast, the GARCH volatility forecasts decay slowly towards the unconditional level, as expected from a stationary GARCH process.

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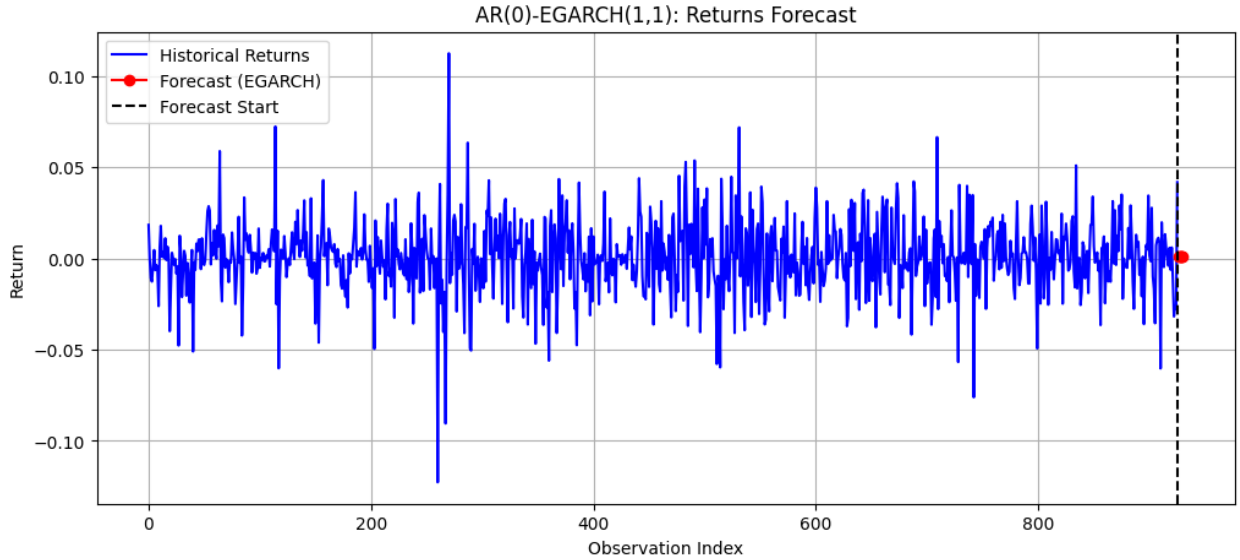

Figure 6: EGARCH returns forecast.

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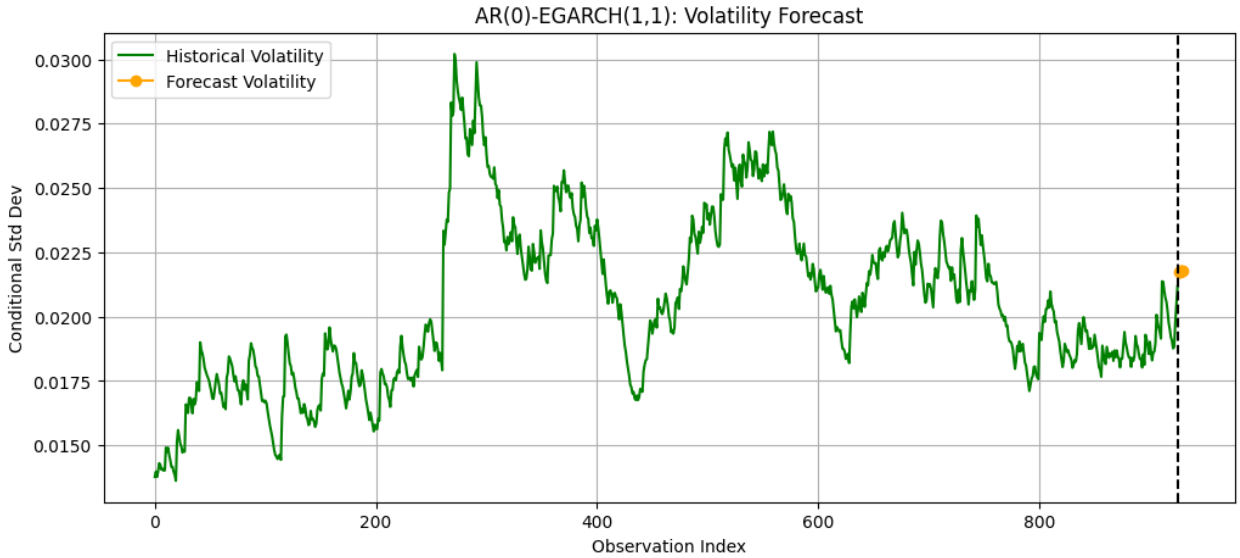

Figure 7: EGARCH volatility forecast.

Figure 8: Forecasts from the EGARCH model. The vertical dashed line marks the start of the forecast horizon.

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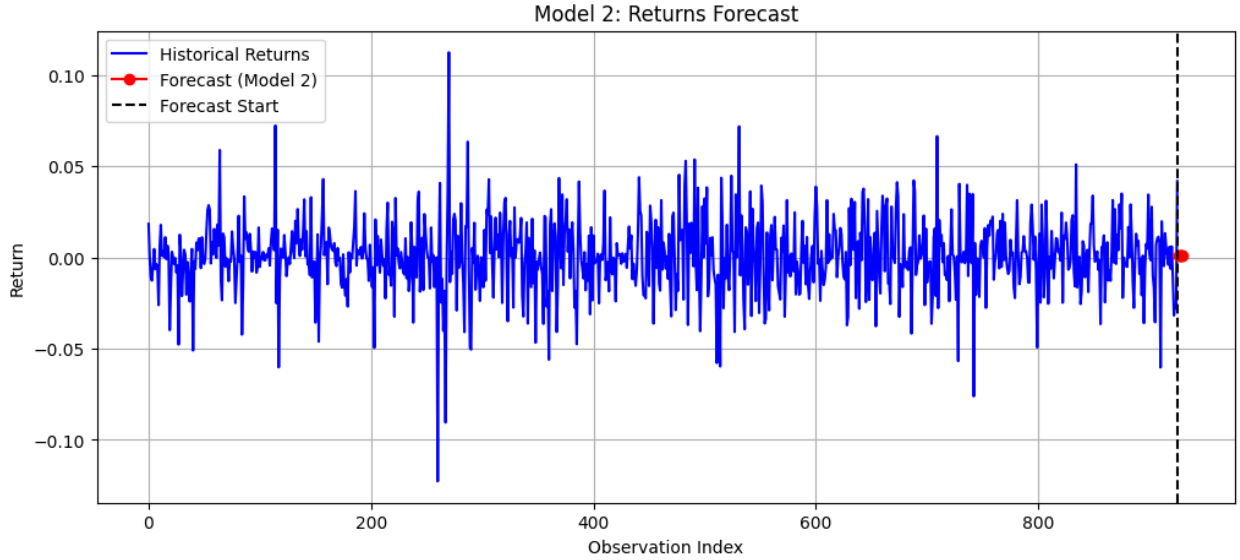

Figure 9: GARCH returns forecast.

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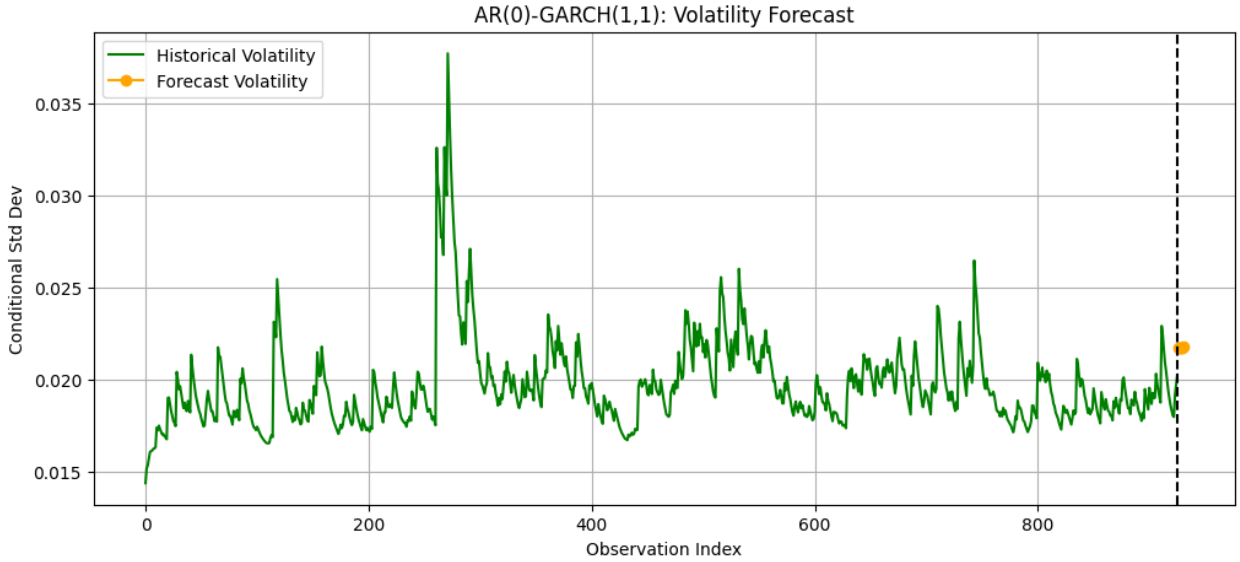

Figure 10: GARCH volatility forecast.

Figure 11: Forecasts from the GARCH model.